

OMFS 553

DIGITAL IMAGING

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Introduction

Efficiency and Instant Viewing

- Digital imaging allows immediate viewing of dental images, reducing waiting time and improving diagnostic decisions.

Radiation Dose Reduction

- Digital sensors require less x-ray exposure, supporting the ALARA principle for patient safety.

Environmental Benefits

- Digital imaging eliminates chemical processing and lead waste, reducing environmental impact of dental radiography.

Improved Communication and Storage

- Digital images are easy to store, share, and enhance patient understanding during consultations.

Learning Goals and Clinical Relevance

Digital Imaging Fundamentals

- Understand digital image formation, receptor types, and their role in clinical decision making.

Image Quality Assessment

- Learn to evaluate contrast, resolution, noise, and exposure for diagnostic accuracy.

Clinical Safety and Usage

- Recognize exposure errors and artifacts to reduce patient dose and ensure effective imaging.

Data Storage and Security

- Understand digital image storage, backup, and privacy to ensure legal compliance.

The Dental Imaging Chain

X-ray Generation and Attenuation

- X-rays generated pass through oral tissues, attenuated by enamel, dentin, bone, and soft tissue based on density and thickness.

Image Receptor Function

- The receptor detects remaining x-ray energy and converts it into electrical or stored signals for image creation.

Digital Image Processing

- Computer systems convert signals into digital images composed of pixels with gray scale values.

Image Display and Quality

- Image display quality and environment impact perception of detail; errors can degrade final image quality.

Transition from Film to Digital Systems

Limitations of Film-Based Imaging

- Film imaging required chemical processing, physical storage, and manual handling, causing inefficiencies and variability.

Advantages of Digital Imaging

- Digital systems provide immediate images, reduce appointment times, and offer consistent, high-quality results.

Environmental and Regulatory Impact

- Digital imaging eliminates chemical waste and lead foil, aligning with environmental safety and regulatory standards.

Integration with Health Records

- Digital radiography integrates with practice software, supporting electronic health records and streamlined workflows.

Defining Digital Radiography

Digital Radiography Basics

- Digital radiography uses electronic detectors and computers to capture and process x-ray images instead of film.

Essential Components

- The system requires an image receptor, a computer with software, and a display device for image viewing and processing.

Image Manipulation Advantages

- Digital images can be adjusted for brightness, contrast, and magnification without retaking the x-ray, improving visualization.

Clinical Practice Considerations

- Digital radiography enhances efficiency but requires proper exposure techniques and should not replace radiographic principles.

Basic Structure of a Digital Image

Digital Image Matrix

- A digital image is composed of a grid called a matrix with rows and columns defining pixel locations.

Pixel Intensity Values

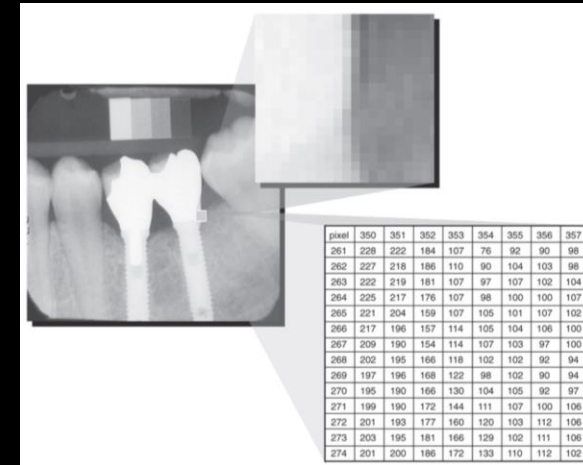
- Each pixel holds a numerical value corresponding to signal intensity, shown as a specific gray shade.

Image Quality Factors

- Image quality depends on pixel number, size, and the available range of gray values for detail display.

Digital Image Advantages

- Digital images can be manipulated, copied without quality loss, and quantitatively analyzed unlike traditional photographs.



Pixels and Gray Values

Pixel and Gray Value Basics

- Pixels are the smallest units of a digital image. each pixel has an intensity value ranging from 0 to 255, representing brightness levels.

X-ray Absorption and Image Brightness

- Dense structures attenuate more x-rays and appear lighter; less dense areas appear darker in the image.

Image Processing Limitations

- Pixel values are fixed at exposure; software can adjust brightness but cannot create new image data.

Importance of Exposure and Positioning

- Correct exposure and positioning ensure accurate pixel data and reduce noise or pixelation in images.

0=BLACK
255=WHITE



Advantages of Digital Imaging

Reduced Radiation Exposure

- Digital imaging uses more sensitive receptors, allowing lower x-ray doses for safer diagnostics.

Speed and Efficiency

- Images are instantly available after exposure, improving workflow and reducing patient time.

Easy Storage and Sharing

- Digital images are stored electronically, simplifying retrieval and sharing without quality loss.

Environmental Benefits

- Eliminates chemical processing and lead waste, promoting environmentally friendly dental practices.

Disadvantages and Limitations

High Initial Costs

- Setting up digital imaging systems requires significant investment in sensors, software, and computers.

Fragility of Digital Sensors

- Digital receptors are more delicate than film and can be damaged by mishandling or improper cleaning.

Technological Dependence

- Digital imaging relies heavily on hardware, software, and network stability, which can cause workflow disruptions.

Training and Experience Needed

- Effective use of digital imaging systems requires proper training and experience for clinicians.

How Digital Images Are Created

X-ray Exposure and Interaction

- X-rays pass through oral tissues and interact based on tissue density and thickness, affecting radiation reaching the receptor.

Radiation Detection and Conversion

- Image receptors convert radiation into electronic signals, using solid-state sensors or photostimulable phosphor systems.

Image Processing and Display

- Computers organize pixel values and gray levels to create digital images shown on monitors for diagnosis.

Overview of Digital Image Receptors

Solid-State Sensors

- Solid-state sensors like CCD and CMOS deliver near-instant images and come in wired or wireless forms.

Photostimulable Phosphor Plates

- PSP plates mimic film size and flexibility but require scanning to produce digital images.

Comparison Factors

- Each receptor type varies in image quality, workflow efficiency, cost, and patient comfort.

Solid-State Sensors in Practice

Digital Radiography Advantages

- Solid-state sensors provide fast image capture and high-quality digital images for dental diagnostics.

Sensor Physical Characteristics

- These sensors are thicker and less flexible than traditional film, affecting patient comfort and positioning.

Durability and Handling

- Solid-state sensors are durable but sensitive to rough handling, requiring careful use to prevent damage.

Clinical Applications

- Rapid image availability makes these sensors ideal for immediate feedback during procedures like endodontics and implants.

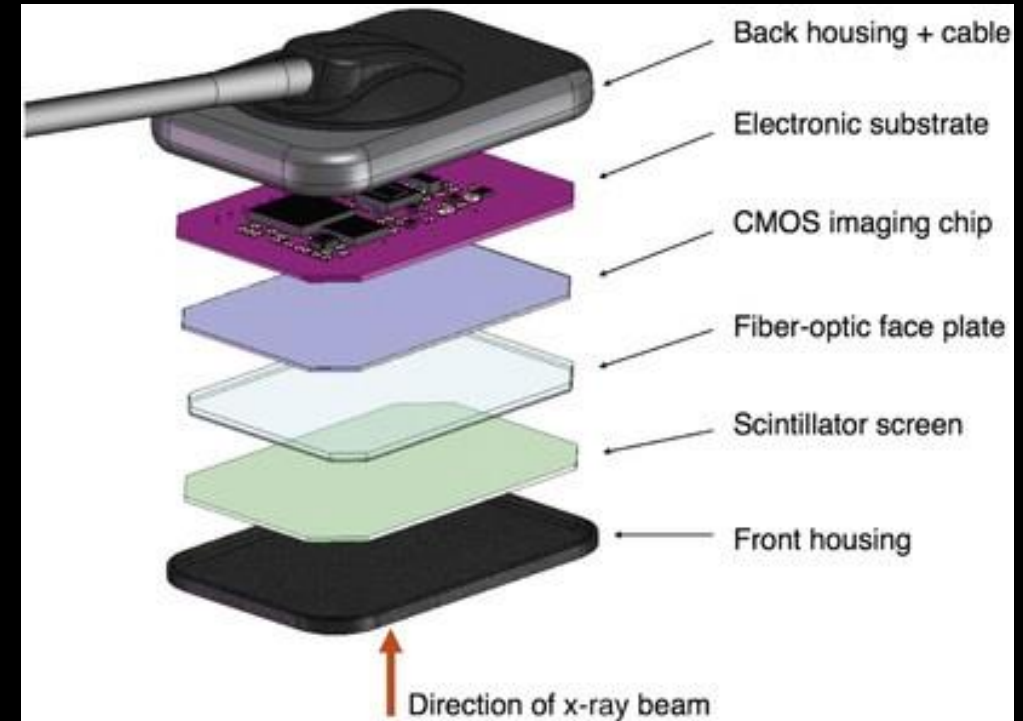
Charge Couple Devices (CCD)

- Solid-state digital sensor
- Converts x-rays into electrical charge
- Charges move pixel-to-pixel to one output
- Produces high-quality images
- Used in early digital radiography



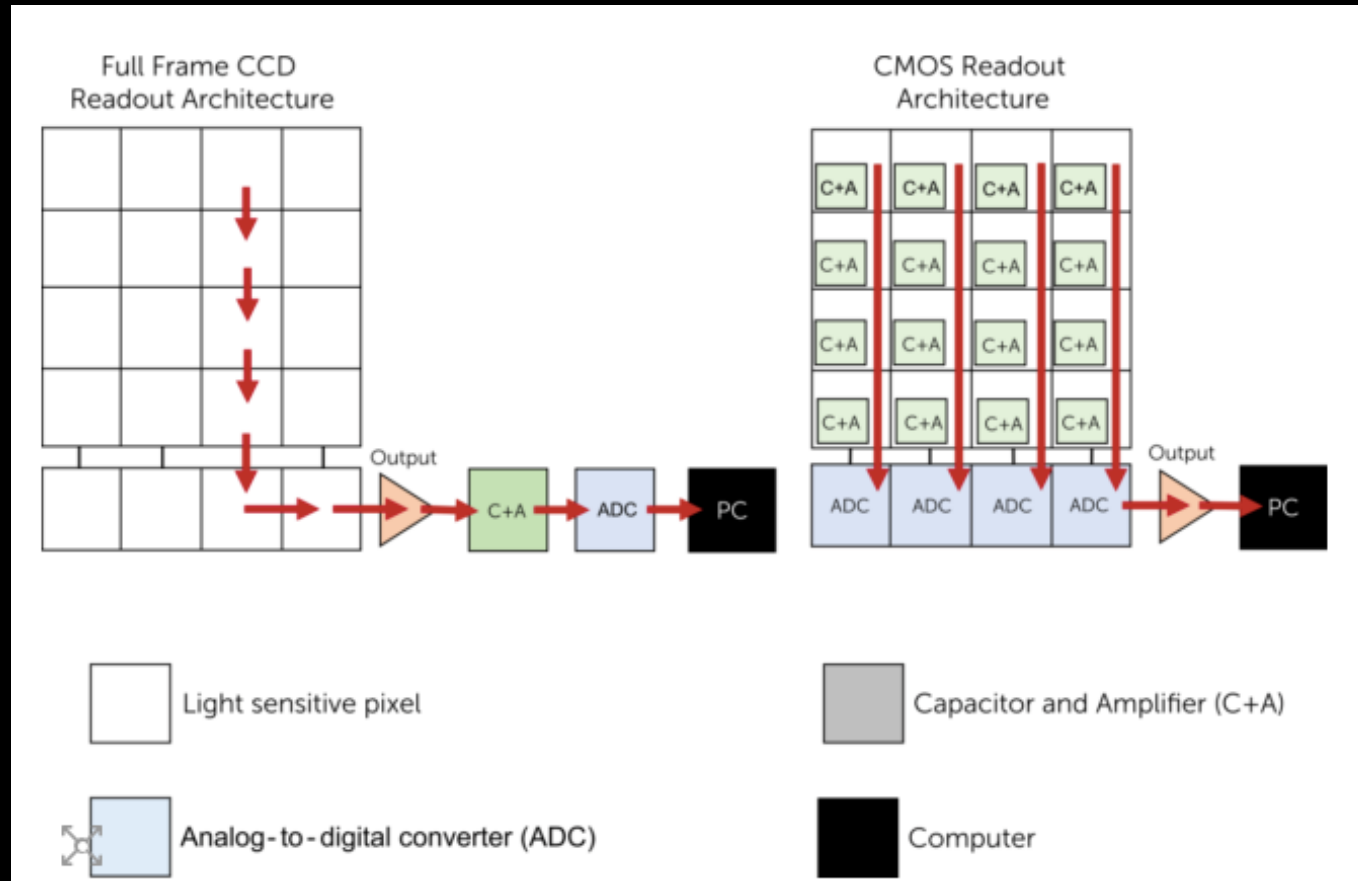
Complementary Metal Oxide Semiconductors (CMOS)

- Solid-state digital sensor
- Each pixel has its own transistor
- Signal read directly from each pixel
- Lower power use and lower cost
- Most common sensor today



Exploded View of CMOS

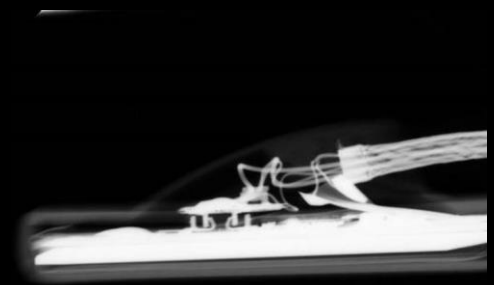
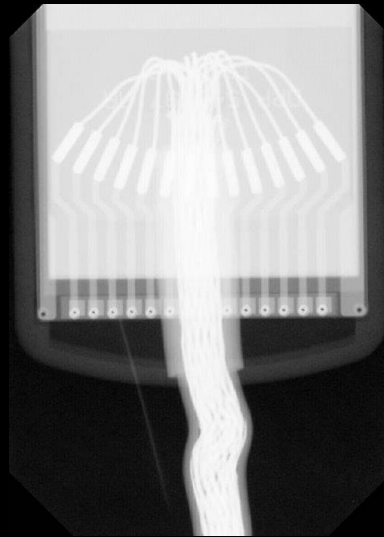
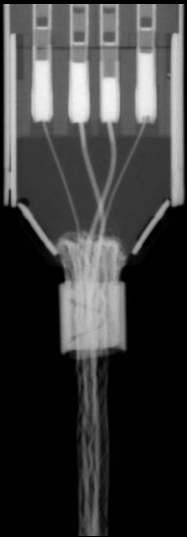
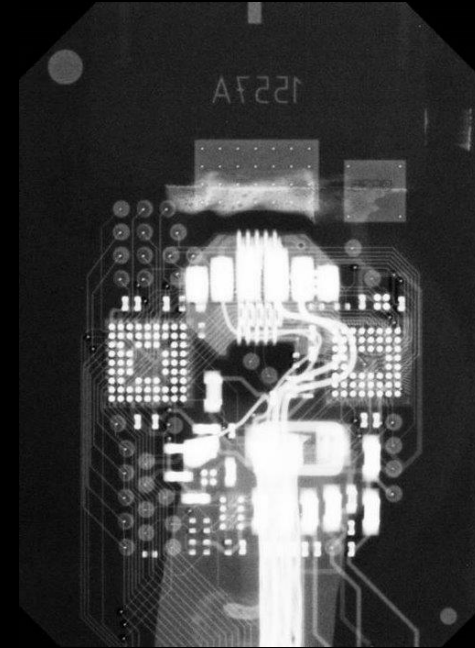
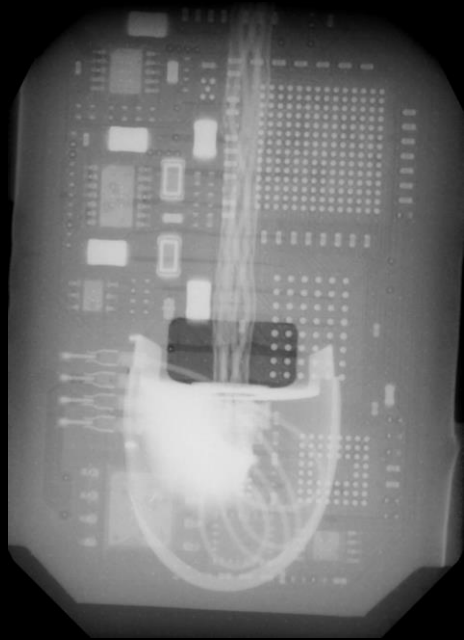
row-by-row



each pixel connected
to separate transistor

conversion of photonic signals into electron packets that are then transferred to a common output structure to convert the electric charge into a voltage

charge-to-voltage conversion on a CMOS imager occurs in each pixel, thereby allowing most functions to be integrated directly into the chip



Photostimulable Phosphor Systems

Energy Storage and Release

- Flexible phosphor plates store x-ray energy and release it as light when scanned by a laser after exposure.

Image Detection and Conversion

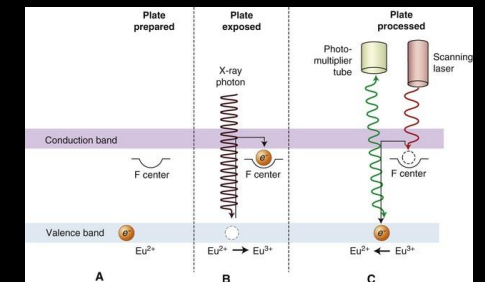
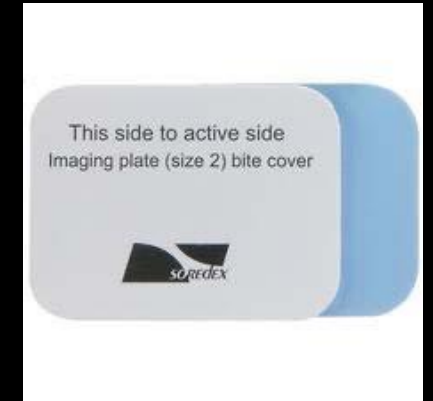
- The emitted light from the plate is detected and converted into a digital image for diagnostic use.

Patient Comfort and Handling

- PSP plates are wireless and film-sized, offering patient comfort but require careful handling to avoid artifacts.

Exposure Latitude and Risks

- Wide exposure latitude of PSP systems allows forgiving imaging but needs management to prevent overexposure.



Clinical Comparison of Intraoral Imaging Alternatives

Imaging Step	Film	CCD/CMOS	PSP
Receptor preparation	None	1. Place protective plastic sleeve over receptor envelope 2. Receptor must be connected to computer and patient-identifying information entered for acquisition/archiving software	1. "Erase" plates 2. Package plates in protective plastic
Receptor placement	1. Numerous generic film-holding devices are available 2. Film may be bent to accommodate anatomy	1. Specialized receptor holder specific for manufacturer's receptor may limit options 2. Receptor inflexibility and bulk limit placement options 3. Receptor cable must be carefully routed out of patient's mouth 4. Patient discomfort more likely than with film or PSP	1. Many receptor holders used for film may be adapted for PSP plates 2. Bending of receptor may irreversibly damage it
Exposure	1. Simple exposure	1. Computer must be activated before exposure	1. Simple exposure

Clinical Comparison of Intraoral Imaging Alternatives

Imaging Step	Film	CCD/CMOS	PSP
Processing	1. Dark, light-safe environment in form of darkroom or daylight loader required 2. Processor chemistry must be prepared or replenished 3. Chemical temperature must be warmed, or processing time must be adjusted to accommodate temperature 4. Films must be removed from wrapper; lead foil must be separated for recycling	1. Image acquisition and display almost immediate	1. Dimly lit environment to prevent loss of image information 2. Processor must be programmed with patient information so that the images are stored properly. Alternatively, plates can be preprogrammed with patient information using RFID 3. Processor must be programmed with detector information so that the images are preprocessed 4. Protective wrapper must be removed from plates
Display preparation	1. Films may be placed in a variety of film mounts 2. Mounts must be labeled with patient-identifying information	1. Software may be configured to place image in appropriate position in digital mount when exposures are made in a predetermined sequence; otherwise images must be individually placed in the mount	1. Images must be individually placed in mount 2. Images may have to be digitally rotated to achieve proper orientation
Display	1. A room with subdued lighting and a masked viewbox are optimal 2. Any light source (including the operatory window or ceiling light) permits quick evaluation of the image	The same considerations apply to all digital receptor types: 1. A room with subdued lighting is optimal for interpretation activities 2. Computer with appropriate software and a display with adequate resolution are necessary; viewing is restricted to the location of the computer 3. Laptop computers increase flexibility of computer placement 4. Size of display restricts the numbers of images that may be viewed simultaneously; more time is required to open/close or expand/contract images when interpreting a series of images	
Image duplication	1. Quality of duplication is always inferior to original and is sometimes nondiagnostic	1. Electronic copies may be stored on various media without loss of image quality 2. Output on film or paper is inferior and is often nondiagnostic unless appropriate combinations of high-resolution printers and papers or film are used	

Image Quality Factors

Contrast and Spatial Resolution

- Contrast resolution allows differentiation of tissue densities; spatial resolution reveals fine anatomical details.

Noise in Radiography

- Noise causes unwanted image variations that can obscure diagnostic information and reduce clarity.

Detector Latitude and Sensitivity

- Detector latitude defines exposure range for usable images; sensitivity indicates detector efficiency to radiation.

Impact of Technique and Equipment

- Proper technique, positioning, and equipment use minimize image degradation and improve diagnostic outcomes.

Contrast Resolution

Definition of Contrast Resolution

- Contrast resolution measures how well an imaging system distinguishes small differences in x-ray attenuation.

Importance in Dental Radiography

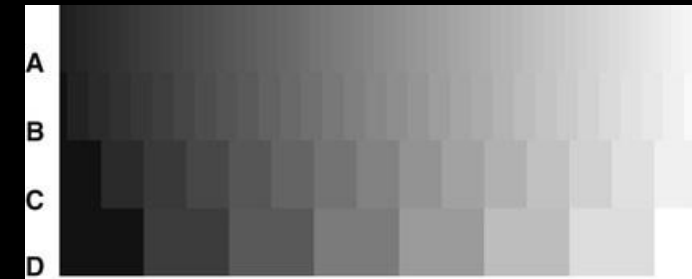
- It is critical for detecting subtle dental issues like early caries and bone loss in radiographs.

Limitations of Display and Vision

- Monitor and human vision constraints reduce the number of gray levels that can be effectively perceived.

Role of Image Processing

- Image processing enhances contrast but may cause artifacts or hide details if overdone.



- A) 6 bits/pixel = 64 gray levels.
- B) 5 bits/pixel = 32 gray levels.
- C) 4 bits/pixel = 16 gray levels.
- D) 3 bits/pixel = 8 gray levels.

Film > Digital

Spatial Resolution

Definition of Spatial Resolution

- Spatial resolution measures the ability to distinguish fine detail, often in line pairs per millimeter.

Factors Affecting Resolution

- Pixel size, detector design, and focal spot size influence spatial resolution in imaging.

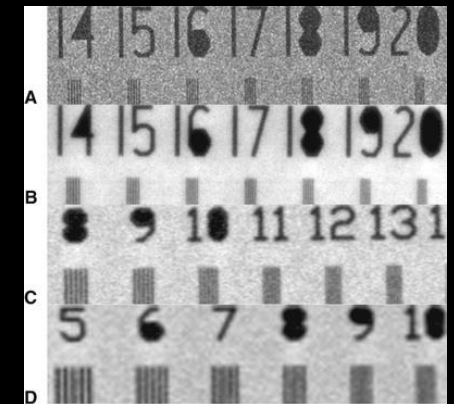
Limitations in Clinical Settings

- Patient movement and system noise limit clinical resolution despite technological advances.

Digital vs Film Resolution

- Film offers very high resolution; digital imaging provides enough detail for diagnostics but magnification reveals pixel structure.

Test Phantom With Various Receptors



- (A) Kodak (Atlanta, GA) InSight film.
- (B) Trophy RVGui (Kodak) high- resolution CCD
- (C) Gendex (Hatfield, PA) DenOptix PSP scanned at 600 dots per inch (DPI).
- (D) Gendex DenOptix PSP scanned at 300 DPI.

Film > CCD/CMOS > PSP

Direct digital characteristics-spatial resolution

Spatial Resolution	lp/mm
Human Eye	6.0 (20+mag)
PSP	7.0-18.0
CMOS	12.5-18.0
CCD	12.5-26.3
Film	20.0

Film $\geq$ CCD = CMOS > PSP

Intraoral sensors only

Detector Latitude

Definition of Detector Latitude

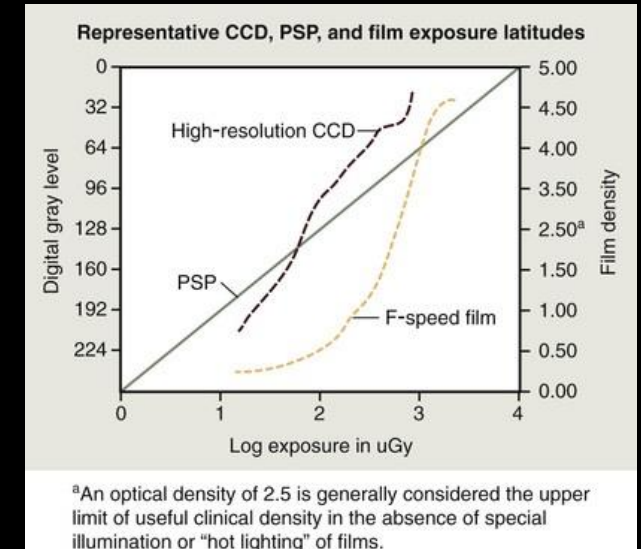
- Detector latitude refers to the range of x-ray exposures producing diagnostic-quality images, capturing both dense and less dense tissues.

Comparison of Detector Types

- PSP systems have wider latitude than solid-state sensors or film, making them more tolerant of exposure errors.

Clinical Implications

- Wide latitude can mask overexposure, increasing patient dose risk; clinicians must understand latitude to optimize exposures.



PSP > CCD/CMOS > Film

Detector Sensitivity and Dose

Definition of Detector Sensitivity

- Detector sensitivity measures how efficiently receptors respond to x-ray exposure, impacting image quality and safety.

Importance for Patient Safety

- Higher detector sensitivity reduces required radiation dose, enhancing patient safety during imaging procedures.

Digital vs Film Systems

- Digital detectors typically allow dose reduction compared to film, but sensitivity varies by type and manufacturer.

Dose Optimization Awareness

- Clinicians should avoid increasing exposure to improve image appearance due to post-processing capabilities, ensuring dose optimization.

Comparison of Physical Properties

Feature	Technical Comment	Clinical Comment
Spatial resolution	Intraoral systems: Film > CCD = CMOS > PSP Panoramic systems: Film = CCD = PSP Cephalometric systems: Film > CCD = PSP	Limits of resolution for digital systems are readily appreciated when magnifying these images. With magnification, a "blocky" or "pixelated" appearance is evident. Resolution of panoramic systems is limited by mechanical motion to about 5 lp/mm
Exposure latitude	PSP ≧ CCD ≥ CMOS > film	Because of the wide latitude of PSP and the automatic brightness and contrast "optimization" by image acquisition software, use of more x-ray exposure than is necessary is possible
Receptor dimensions	For equivalent imaged area, film = PSP < CCD = CMOS	"Active area" of CCD and CMOS receptors is smaller than surface area because of other electronic components within the plastic housing
Time for image acquisition	CCD = CMOS ≪ PSP = film	Rapid image acquisition may be important for endodontic procedures or during implant placement
Image quality	Subjective quality is best with film when carefully exposed and well processed	Digital and film imaging are not significantly different when used for common diagnostic tasks
Image adjustment/processing	Improves appearance of digital image	Takes time; may not improve diagnostic performance
Cost	Initial costs of digital systems are greater than those of film. Subsequent costs vary greatly depending on receptor wear and tear or abuse	Manufacturers' estimates of life expectancy of reusable receptors may be overly optimistic
Reliability	Mechanical problems affect digital PSP and film systems. Software reliability varies greatly among manufacturers. Changes in unrelated computer components and software can cause digital systems to malfunction	Digital systems fail when problems occur with receptors during image acquisition or with computers during image processing, archiving, and display
Image storage and retrieval	Data backup and security is critical for digital systems	Films can be misfiled and lost or be damaged by poor storage conditions. Digital data can be lost as a result of failures in power supplies or storage media and operator error
Transmitting images to others	Rapidly done with digital images	Facilitates communication between colleagues or with insurance companies

Digital Image Viewing Environment

Monitor Quality Requirements

- Monitors must have sufficient brightness, contrast, and resolution for clear dental image display. This ensures diagnostic details are visible.

Ambient Lighting Conditions

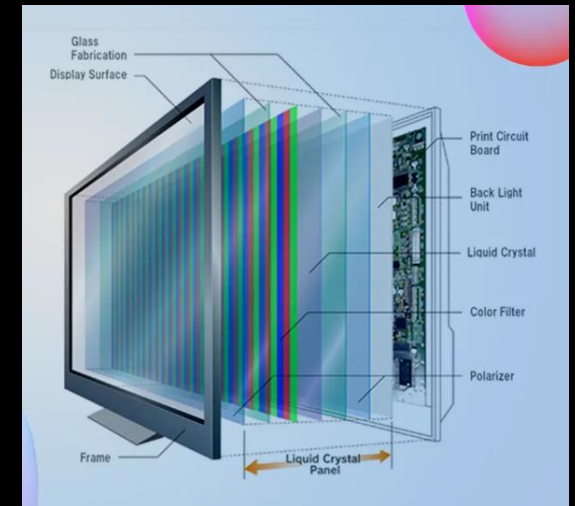
- Ambient lighting should be subdued to minimize glare and reflections that can interfere with image viewing accuracy.

Screen Positioning and Viewing Angle

- Proper screen positioning and viewing angle are essential to avoid distortion and ensure accurate image perception.

Impact on Diagnostic Accuracy

- Poor viewing conditions can obscure subtle image details, reducing the accuracy of dental diagnoses.



Types of Displays Used in Dentistry

Common Display Types

- LCD and LED flat-panel displays are commonly used in dental practices for viewing digital radiographs with adequate brightness and contrast.

Medical-Grade vs Consumer Displays

- While specialized medical-grade monitors exist, high-quality consumer displays generally meet dental imaging needs effectively.

Display Considerations

- Important factors include screen size, resolution, and image stability over time to ensure accurate and consistent image presentation.

Hard Copies and Printing

Need for Hard Copies

- Despite digital viewing, hard copies remain essential for referrals, education, and record keeping in clinical settings.

Image Quality Reduction

- Printing reduces image quality, especially contrast resolution, due to printers' inability to reproduce true grayscale tones.

Printer Types and Limitations

- Inkjet and dye-sublimation printers are commonly used, each with specific advantages and inherent limitations in image reproduction.

Clinical Use Caution

- Clinicians should use printed images cautiously as they are generally inferior to on-screen images for diagnostic purposes.

Image Processing Concepts

Digital Image Modification

- Image processing involves digital operations that alter images after acquisition to enhance visual quality.

Automatic Processing Steps

- Automatic processes like noise correction and grayscale leveling are performed without user intervention.

User-Controlled Adjustments

- Users can control brightness, contrast, and sharpness to customize image appearance.

Clinical Interpretation Caution

- Processing improves visuals but does not add diagnostic data; clinicians must avoid overinterpretation.

Enhancement Versus Diagnosis

Purpose of Image Enhancement

- Enhancement improves visual appeal but does not guarantee diagnostic accuracy or reliability.

Risks of Over-Enhancement

- Excessive sharpening or contrast can create artifacts and highlight irrelevant features, misleading diagnosis.

Diagnostic Best Practices

- Clinicians should rely on anatomical knowledge and multiple views, not solely on heavily processed images.

Brightness and Contrast Adjustment

Function of Brightness

- Brightness controls the overall lightness or darkness of an image affecting image visibility.

Role of Contrast

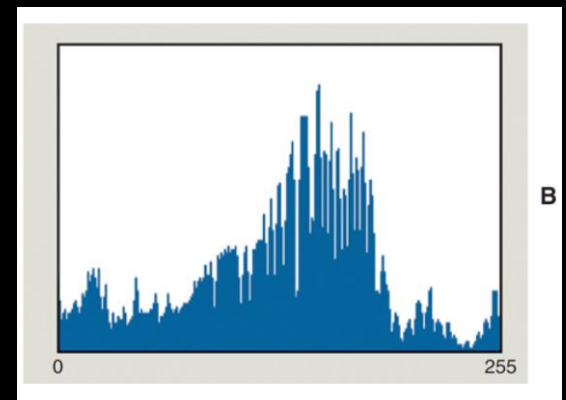
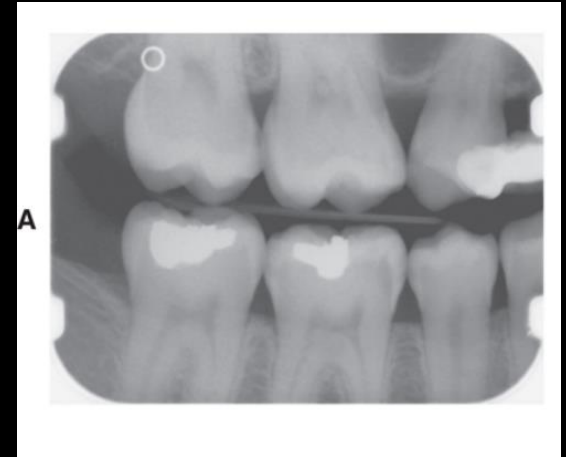
- Contrast controls the difference between light and dark areas, enhancing structural detail visibility.

Importance of Proper Use

- Improper brightness or contrast can obscure details or exaggerate noise in radiographs.

Histogram as Adjustment Guide

- Histograms show gray value distribution and help guide accurate brightness and contrast adjustments.



Sharpening and Smoothing Filters

Purpose of Filters

- Sharpening emphasizes edges, enhancing boundaries; smoothing reduces noise and random variations in images.

Diagnostic Impact

- Filters improve visual appeal but have limited proven impact on diagnostic accuracy in clinical imaging.

Potential Risks

- Over-sharpening can create false edges; over-smoothing may blur critical details, leading to misinterpretation.

Colorization and Inversion

Colorization Technique

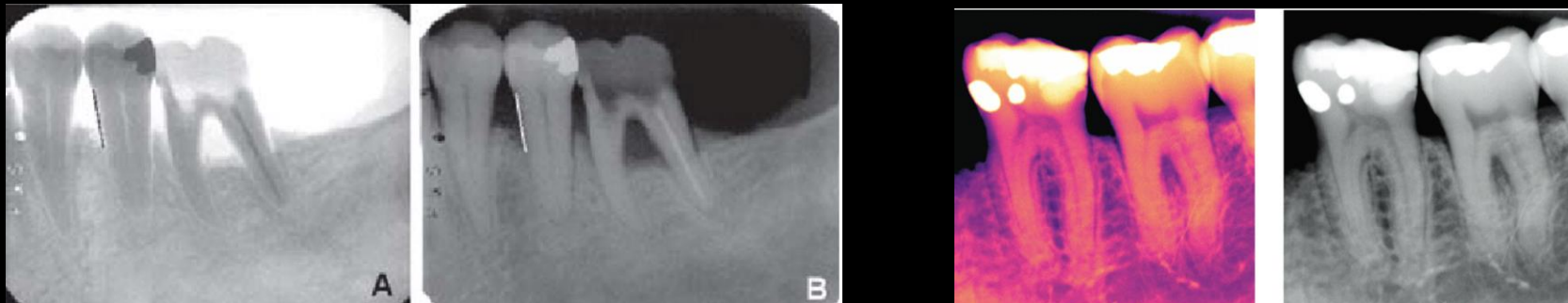
- Colorization assigns artificial colors to grayscale values, altering the traditional radiograph appearance.

Grayscale Inversion

- Inversion reverses black and white tones, creating a photographic negative effect on radiographs.

Diagnostic Preference

- Clinicians prefer conventional grayscale images as they support accurate anatomical interpretation and diagnosis.



Digital Subtraction Radiography

Principle of Digital Subtraction

- This technique compares two radiographic images by subtracting unchanged structures to highlight subtle differences.

Detection of Subtle Changes

- Digital subtraction radiography reveals subtle changes such as bone loss not easily seen on standard images.

Technical Challenges

- Accurate image subtraction requires identical positioning and exposure, making routine use challenging.

Research Applications

- Due to technical demands, subtraction radiography is mainly used in research rather than daily clinical practice.

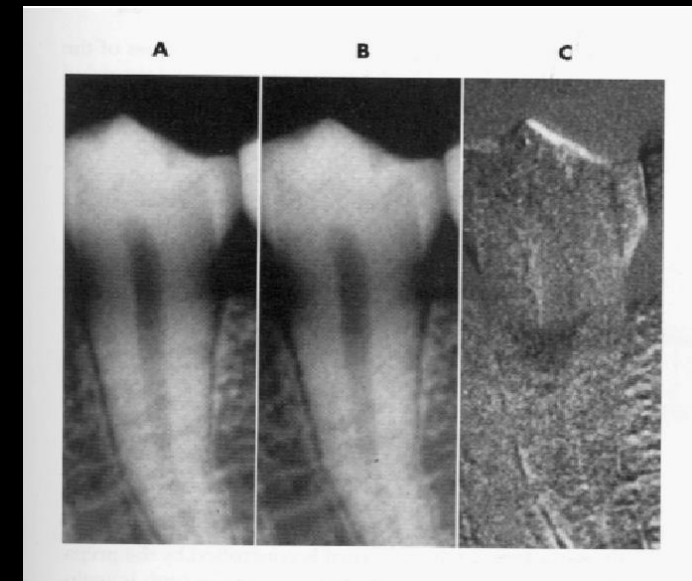


Image Analysis and Measurement

Measurement Tools in Imaging

- Digital tools estimate distances, angles, and densities vital for dental procedures like implant planning and working length estimation.

Importance of Calibration

- Accuracy of image measurements depends on proper calibration and maintaining consistent imaging geometry.

Automated Analysis Limitations

- Automated methods like computer-assisted caries detection are emerging but still show limited reliability in clinical use.

Clinical Judgment Importance

- Clinicians should use image analysis tools as aids and not replace their expert clinical judgment.



Image Storage and Backup

Secure Image Storage

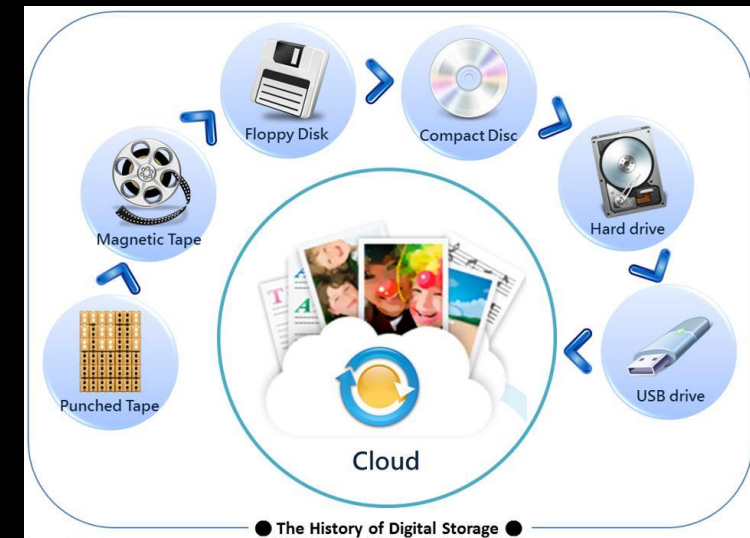
- Digital radiographic images require secure storage to protect patient data and ensure accessibility when needed.

Backup Importance

- Regular backups prevent data loss caused by hardware failure or human error, ensuring data reliability.

Compliance and Responsibilities

- Proper storage and backup practices are essential for legal compliance and maintaining continuity of patient care.



Privacy and Security Considerations

Privacy Regulations Compliance

- Digital images must comply with privacy laws like HIPAA to protect patient information effectively.

Secure Access Controls

- Access to patient images should be restricted to authorized personnel only to maintain confidentiality.

Data Encryption and Transmission

- Encrypting images and secure data transmission prevent unauthorized interception and breaches.

Security Awareness Importance

- Awareness of security considerations helps prevent breaches and maintain patient trust.

Image Compression Basics

Purpose of Compression

- Image compression reduces file size to enhance storage efficiency and speed up data transmission.
- **Lossless Compression**
 - Do not discard image data
 - Exact copy of image is reproduced
 - Max compression rate upto 3:1
 - Gif, raw, bmp, png
- **Lossy Compression**
 - Discards image data, use with caution
 - Does not affect diagnostic quality
 - Compression rates
 - Caries 12:1-14:1, endo file 25:1, pano 28:1
 - Jpeg, mpeg, mp3, wma, aac

DICOM and System Compatibility

DICOM Standard Overview

- DICOM standard enables sharing and viewing of digital medical images across various systems and manufacturers.

Interoperability and Integration

- DICOM compliance supports interoperability and integration with other medical record systems for seamless data access.

Reduced Compatibility Issues

- Using DICOM-compliant systems reduces compatibility problems and facilitates collaboration among healthcare specialists and institutions.



Clinical Handling of Digital Receptors

Handling Solid-State Sensors

- Solid-state sensors must be handled carefully to avoid bending or dropping, ensuring long equipment life and image quality.

Protecting PSP Plates

- PSP plates require protection from light exposure, scratches, and harsh disinfectants to maintain image integrity.

Proper Placement and Training

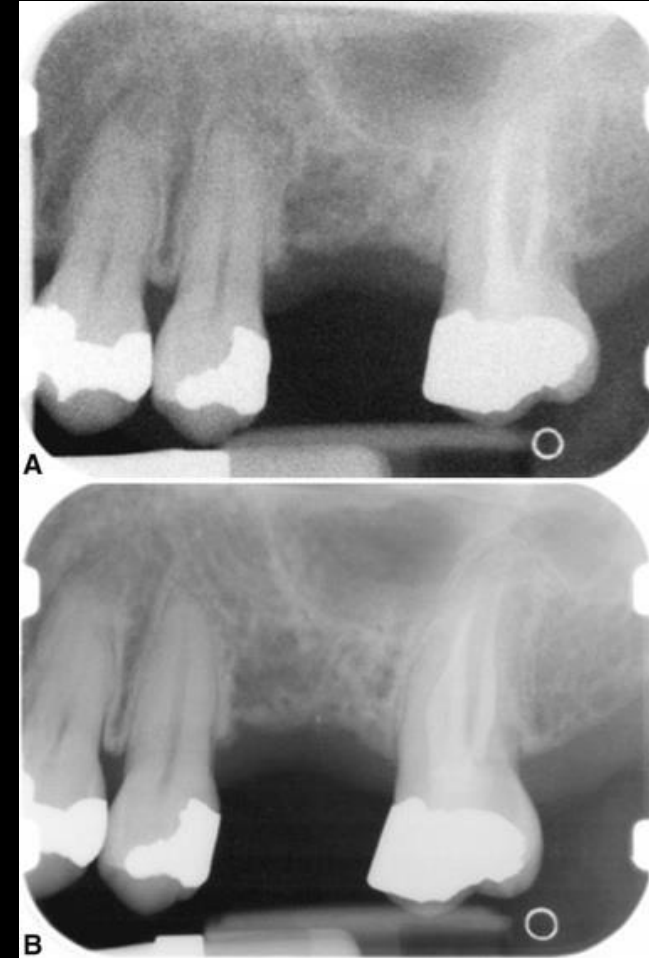
- Correct receptor placement improves image quality and patient comfort; staff training reduces damage and retakes.



Common Problems

Noisy images

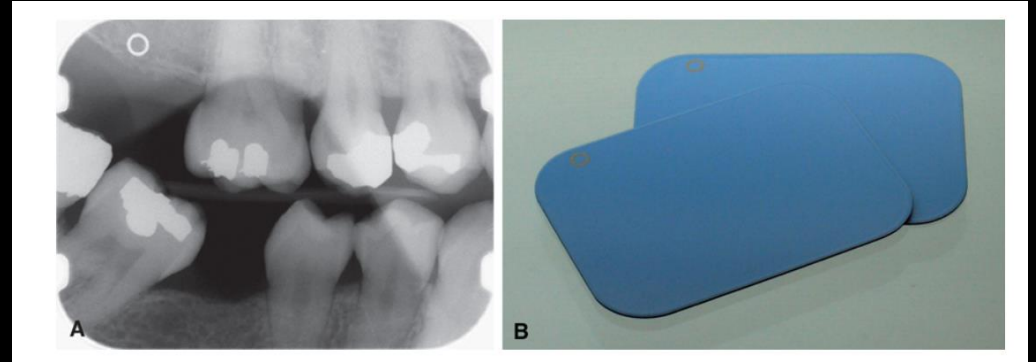
- Although the brightness of these images has been adjusted to display similar average gray values, notice the noisy appearance of the underexposed periapical radiograph (Fig. A, 0.032 s) compared to the properly exposed radiograph (Fig. B, 0.32 s)



Common Problems

Nonuniform image density

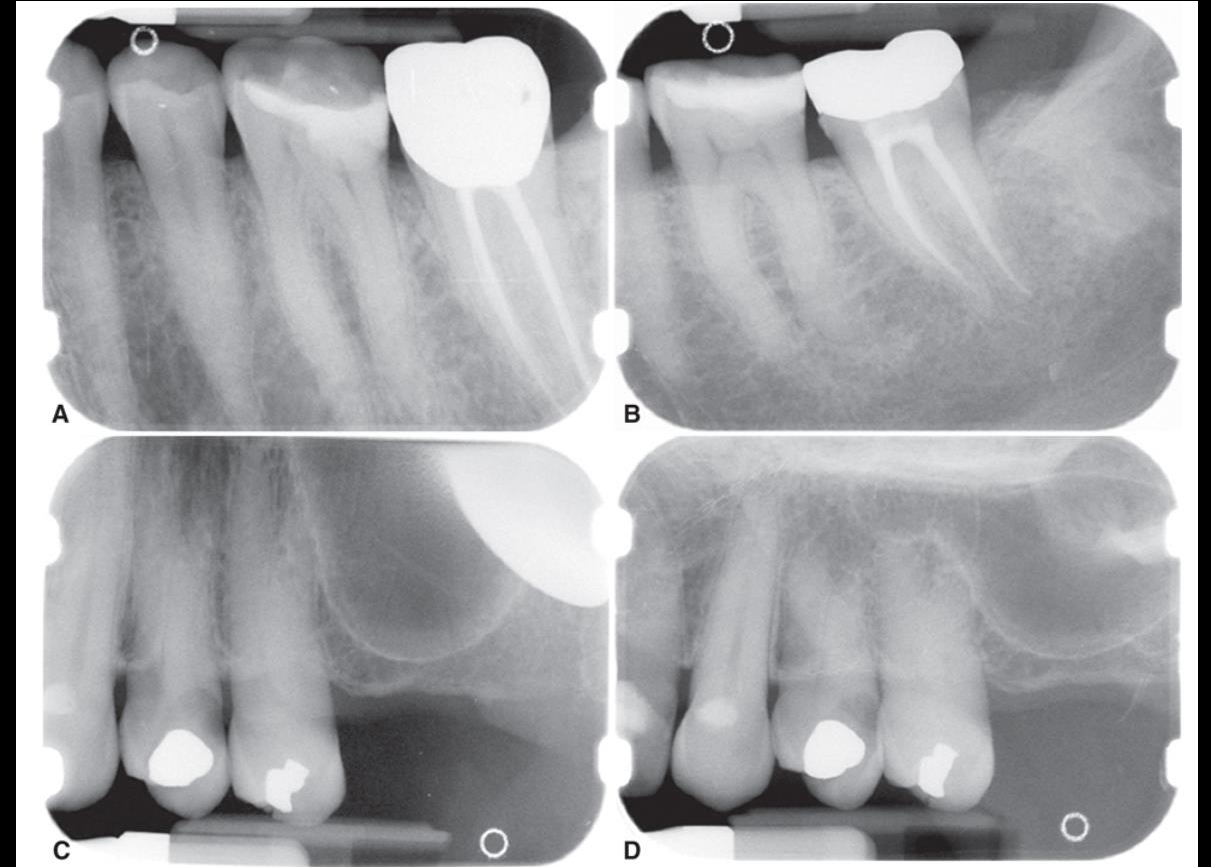
- Partial exposure of PSP plates to excessive ambient light before scanning results in nonuniform image density (A). This happens when plates are overlapped while exposed to ambient light (B)



Common Problems

Distorted images

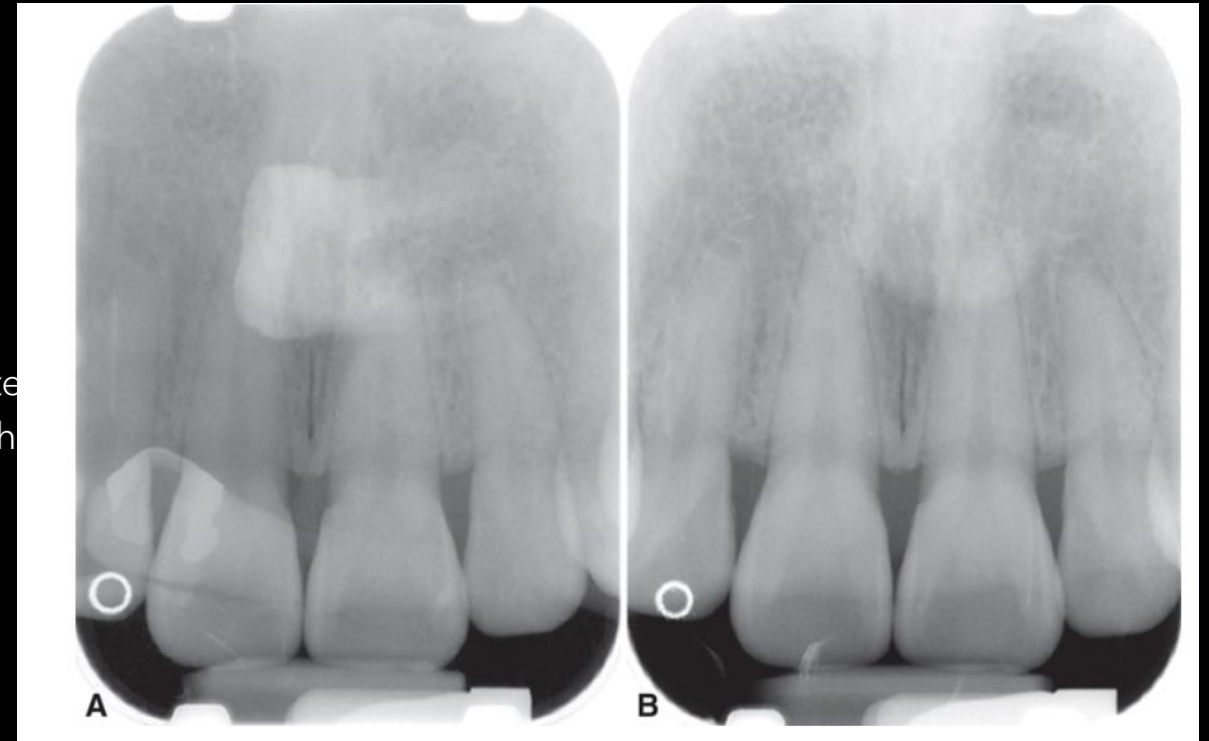
- Bending of PSP plates during intraoral placement: moderate bending (Fig. A), retake of image A (Fig. B), severe bending (Fig. C), and retake of image C (Fig. D)



Common Problems

Double images

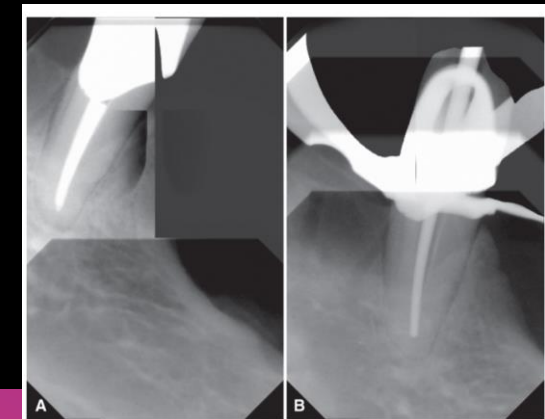
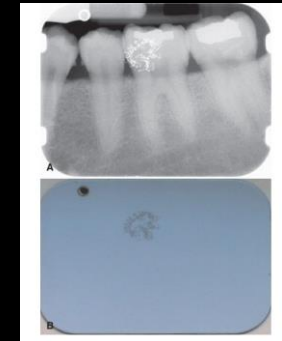
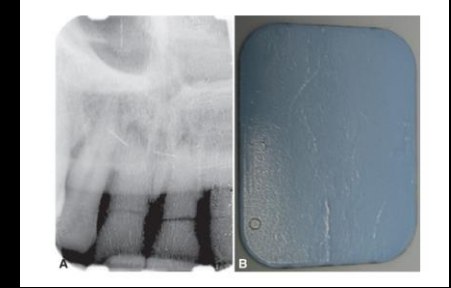
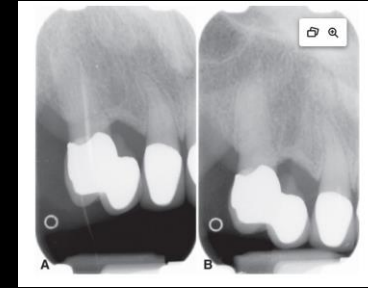
- PSP double image on incisor periapical radiograph resulting from incomplete erasure of previous image of posterior periapical region (Fig. A) and retake of image (Fig.B).
- Another example of a double image resulting from incomplete erasure of a PSP receptor: posterior periapical radiograph with double image (Fig. A), and retake of image A (Fig.B)



Common Problems

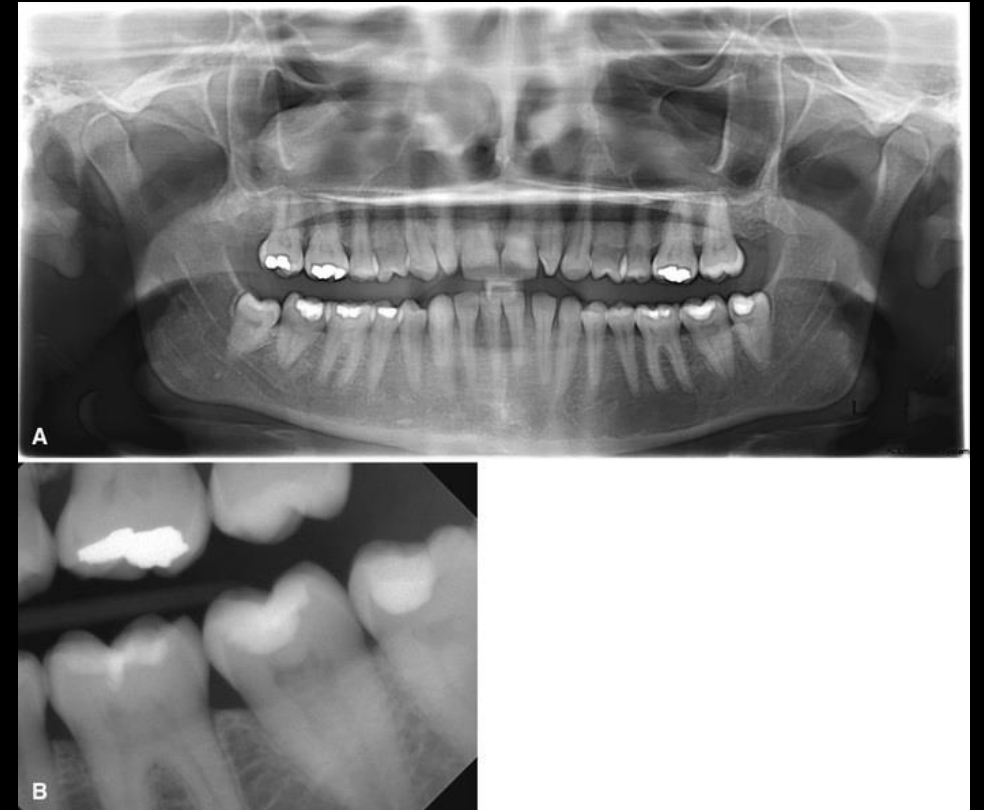
Damaged image receptors

- Scratched phosphor surface mimicking root canal filling (Fig. A) and retake of image (Fig. B).
- Image artifacts resulting from excessive bending of the PSP plate (Fig. A). Excessive bending has resulted in permanent damage to the phosphor plate (Fig. B).
- PSP circular artifact as a result of plate damage (Fig. A). The plate shows localized swelling of the protective coating as a result of ethylene oxide gas used to disinfect the plates (Fig. B). Most likely a pinhole was present in the protective coating allowing the gas to infiltrate and separate the protective coating from the phosphor layer.
- PSP image artifact resulting from exposure of the plate to liquid disinfectant (Fig. A). The disinfectant caused damage to the protective coating and the phosphor layer (Fig. B). Plates should be protected from harsh disinfectants.
- Malfunctioning CCD sensor resulting from incorrect exposure or misinterpretation of the incoming data by the processing software (Fig. 4.26A and B). The sensor was not damaged.



Common Problems

- IMPROPER USE OF IMAGE PROCESSING TOOLS
- Edge enhancement filter
 - Pano: radiolucencies at restoration edges simulating recurrent caries
 - Follow-up intraoral image



Common Problems

- EFFECT OF IMAGE SCANNING RESOLUTION
 - Scanning at 150 dpi (a): image lacks detail and appears pixelated when magnified
 - Scanning at 300 dpi (b): increased detail through higher resolution

